

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda (EMEA, UTC+3)

PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years building **APIs, backend services and SaaS integrations** for product companies, now focused on **integrating AI into customer-facing features**. Shipped a production conversational assistant that combines **OpenAI** with **Google Vision OCR** behind the WhatsApp Business API, and built an agentic evaluation pipeline on Claude Code with a verification gate ahead of every LLM call. Works across stacks by habit — **Node.js and Python** side by side in the same programme, gRPC and Pub/Sub between services — and led a zero-downtime migration of 20+ production applications to GCP, reporting to the CTO. Comfortable bridging Python-based AI services into an existing SaaS ecosystem.

CORE COMPETENCIES

Python & Node.js Backend Services

REST API Design & SaaS Integrations

LLM Integration (OpenAI, Vision OCR)

Agentic Workflows (Claude Code)

Event-Driven Services (Pub/Sub, Kafka)

GCP & Firebase

Cross-Stack Integration (gRPC, Node↔Python)

Zero-Downtime Migrations

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote · e-commerce SaaS

- Wrote the **Python** service that migrated all file storage from Amazon S3 to Google Cloud Storage as part of a company-wide cloud consolidation.
- Built a real-time two-way sync service between MongoDB and Firestore in **Node.js** on Google **Pub/Sub**, with idempotent, redelivery-safe message handling — keeping two live systems consistent under production traffic.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO; moved all services off AWS, saving **\$5,000/month**.
- Shipped customer-facing features end to end: a 3D visualisation feature (video overlay + ffmpeg) that improved buyer conversion, and Express Shipping (+\$40 revenue per order).

MTailor Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Documented the Firebase SDKs for **Node.js and Python** so fellow engineers could build against the new platform; trained the Ops team on the dashboard and backup procedures.
- Built the migration script that read from MongoDB, applied complex transformation logic, and wrote to Firestore.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Extended **REST APIs** in TypeScript on Firebase Cloud Functions and Express.js; integrated **Morningstar market-data APIs** via scheduled jobs over 2M+ Firestore records to keep customer prices current.
- Added responsive PWA pages in React and Tailwind alongside the backend work.

Mind2matter Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built backends for DeFi applications in **Node.js** with Web3 integration; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech

- Architected an event-driven microservices backend on **Apache Kafka, Docker and Kubernetes** for FIDA Uganda's case-management platform; led a team of 4 developers.
- Built a USSD legal-aid service integrating with the Node.js backend over **gRPC**, with a provider-agnostic contract so multiple providers integrated without bespoke code.
- Delivered a paralegal database for LASPNET Uganda and a cross-platform React Native app with Firebase push notifications.

PROJECTS

Healthcare WhatsApp Assistant LLM in production

Conversational assistant that reads prescription photos and responds to patients inside WhatsApp: OpenAI for dialogue, Google Vision for OCR, orchestrated behind the WhatsApp Business API with media ingestion, dialogue state and platform rate/window limits handled.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

Agentic Evaluation Pipeline AI tooling

Extended an open-source agentic pipeline on Claude Code that scans job portals, verifies each posting live in a headless browser before any LLM scoring, deduplicates sources and generates tailored documents — 200+ evaluations with verification-before-inference as the core design rule.

Claude Code · Playwright · Node.js · YAML

Real-time Pub/Sub Sync Service

Two-way MongoDB↔Firestore integration with idempotent writes and conflict handling, running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB matching users to service providers from messy, user-generated catalog data.

Node.js · MongoDB

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions

AI / LLM: OpenAI API, Google Cloud Vision, Claude Code agentic workflows, prompt and tool orchestration

Messaging & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore

Cloud / Infra: GCP (Firestore, Pub/Sub, Cloud Functions, GCS), Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux

Spoken: English, Kiswahili